Team Details

a. **Team name:** Black Bulls
b. **Team leader name:** Shubham
c. **Problem Statement:** Open Track — AI copilot for channel-sourced retail lending (conversational AI · product advisory · financial-health scoring · default prediction)

Brief about the idea

A channel-aware loan-origination & partner-relationship copilot, **on top of IDBI's existing LOS / BRE / LMS.**

Retail lending is sourced through DSAs & dealers and runs on TAT + trust — so it works all four sides at once.

CUSTOMER

Only suitable, clearly-explained, instant offers — no spam, no black box.

RM / BANK

A queue ranked by risk-adjusted profit, with a reason for every score.

CHANNEL PARTNER (DSA)

Instant decisions, transparent auto-payouts, tiering — the loyalty moat.

RISK / REGULATOR

Explainable, fair, adverse-selection-aware lending with an audit trail.

Opportunities

- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

HOW IT'S DIFFERENT

- **Channel-first.** Models the DSA relationship — adverse-selection, TAT, tiering, churn.
- **Bank-grade economics.** Ranks by P(convert) × (net interest income − expected credit loss).
- **Explainable.** Per-decision reason codes + Reg-B / RBI adverse-action codes.
- **Integrates, not replaces.** An intelligence layer on top of LOS / BRE / LMS.

HOW IT SOLVES THE PROBLEM

- **Speed + fit.** Right lead → right product → right price, instantly, with a pro offer.
- **Lower NPAs.** Flags risky sourcing before disbursal, not after.
- **Partner loyalty.** Transparent, fast, tiered payouts → more & better volume.
- **The flywheel.** Win the partner → volume → data → sharper models.

List of features offered by the solution

CUSTOMER & RM

- ML conversion-propensity queue (risk-adjusted)
- Explainable financial-health score (0–100)
- Deterministic eligibility, rate & EMI engine
- Suitability-first next-best-product advisory
- Editable offer + risk-based pricing in-band
- Premium branded PDF + WhatsApp / Gmail send
- Per-decision reason codes on every score

CHANNEL & PARTNER

- Partner / lead default-risk (adverse-selection)
- Channel-mix ROI + book concentration (HHI)
- Commission + clawback payout engine
- Duplicate / fraud lead detection
- Instant decision (<50 ms) + SLA / TAT clock
- Partner tiering + churn early-warning
- Agentic AI assistant (natural language)

Process flow diagram or Use-case diagram

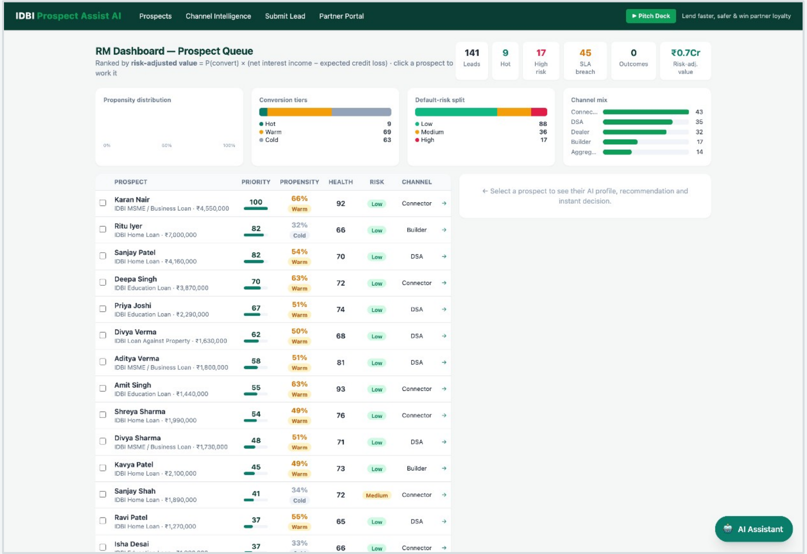
01 Lead submitted → **02** Instant decision → **03** Ranked queue → **04** Offer + price → **05** PDF + send → **06** Outcome logged

↻ **Closed loop.** Outcomes (won / lost / default) retrain the ML models — the compounding data moat.

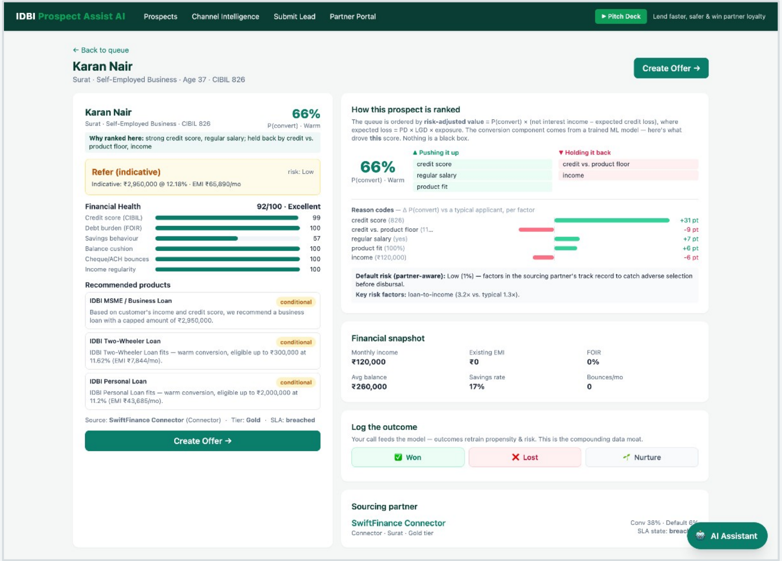
Numbers boundary. The deterministic engine owns every rupee (eligibility, rate, EMI, ECL); the ML owns probabilities; the LLM only phrases. The RM stays in control — every edit re-validates through the engine, so no invalid or non-compliant offer can ever leave.

Wireframes/Mock diagrams of the proposed solution (optional)

RM DASHBOARD — RISK-ADJUSTED QUEUE



PROSPECT PROFILE — REASON CODES



Architecture diagram of the proposed solution

Presentation — FastAPI + Jinja2 + Tailwind (RM dashboard · Partner portal · AI assistant)

Layer 1 — Directives (SOPs in Markdown)

Layer 2 — AI Orchestration (NVIDIA NIM LLM, de-identified)

Layer 3 — Deterministic Engine + ML · owns every number

Eligibility · Rate · EMI	Economics (ECL / RAROC) + pricing	ML: Propensity + Default-risk
Explainability (reason codes)	Commission + Clawback	PDF (WeasyPrint)

Data — products · partners · prospects · history · trained model artifacts (.pkl)

Integration (PoC): CIBIL bureau · KYC / e-consent (DPDP) · LOS / BRE / LMS · NVIDIA AI Enterprise

Technologies to be used in the solution

BACKEND & WEB

Python 3.14 · FastAPI · Uvicorn · Jinja2 · Tailwind CSS · vanilla JS

MACHINE LEARNING & AI

scikit-learn · NumPy / SciPy · joblib · occlusion XAI · NVIDIA NIM LLM

DOCUMENTS & PDF

WeasyPrint · pango / cairo

QUALITY & DEPLOYMENT

pytest (44 tests) · Playwright · Docker · Render / Railway / Cloud Run

Estimated implementation cost (optional)

PROTOTYPE — TODAY

≈ ₹0

Open-source stack
NVIDIA NIM free tier
Free-tier hosting · no licence cost

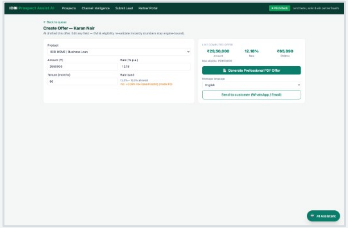
PRODUCTION POC — INDICATIVE (ANNUAL)

- LLM / GPU inference — usage-based
- Cloud hosting + DB + storage — modest
- Integration — LOS / BRE / KYC / CIBIL (one-time)
- Model-ops — monitoring, retraining, governance

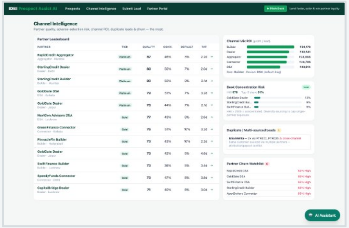
ROI — cost saved (NPA reduction + payout-leakage → 0 + RM selling hours) » cost to run.

Snapshots of the prototype

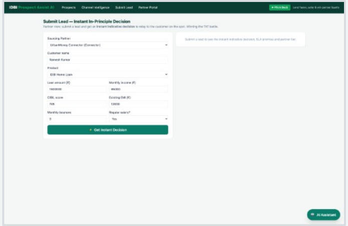
EDITABLE OFFER — LIVE EMI & RISK-BASED PRICING



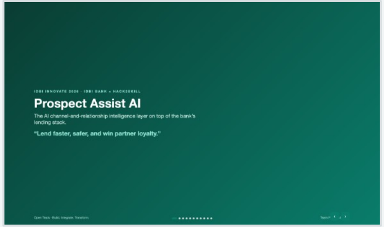
CHANNEL INTELLIGENCE — ADVERSE SELECTION, ROI, HHI



INSTANT DECISION + ADVERSE-ACTION REASON CODES



BUILT-IN PITCH DECK (LIVE NUMBERS)



Prototype Performance report/Benchmarking

MODEL PERFORMANCE — HOLD-OUT, SYNTHETIC (INDICATIVE)

0.69 / 39

CONVERSION PROPENSITY — AUC / GINI

0.81 / 62

PARTNER DEFAULT-RISK — AUC / GINI

SPEED & INTEGRITY

< 50 ms

INSTANT DECISION LATENCY

~1–2 s

LLM REPLY · 9 S TIMEOUT

0.22

FEATURE SHARE — NO LEAKAGE

BUSINESS IMPACT — SYNTHETIC BOOK

₹16.4 Cr

PIPELINE VALUE

₹0.68 Cr

RISK-ADJUSTED VALUE

44 / 44

AUTOMATED TESTS PASSING

Additional Details/Future Development (if any)

01 Integrate

- IDBI sandbox APIs
- CIBIL bureau + KYC
- e-consent (DPDP Act)
- LOS / BRE / LMS hooks

02 Harden

- Retrain on real conv / NPA / TAT
- SHAP + PSI drift monitoring
- Auth / RBAC + audit trail
- Data residency

03 Scale

- Partner mobile portal + WhatsApp
- OTP-gated secure delivery
- Collections & Early-Warning
- NVIDIA AI Enterprise / on-prem

Provide links to your:

- GitHub Public Repository
- Demo Video Link (3 Minutes)
- Final Product Link

GITHUB PUBLIC REPOSITORY

https://github.com/Shubham-33/Hack2Skill-IDBI-Hackathone

FINAL PRODUCT LINK (LIVE PROTOTYPE)

https://prospect-assist-ai-2guf.onrender.com

DEMO VIDEO LINK (3 MINUTES)

[paste your unlisted YouTube / Drive link]

IDBI BANK

IDBI 2026 INNOVATE

Build. Integrate. Transform.

Knowledge Partner aws

Powered by H2S

Technology Partner acc applied cloud computing

Thank you!